Revision Games — 2.2 Problem Solving & Programming

OCR A Level Computer Science (H446) — Component 02, Section 2.2 (2.2.1 Programming techniques · 2.2.2 Computational methods).

A print-and-play pack: **Loop cards (dominoes)**, **Taboo cards**, **Bingo**, and **Quick-fire "Guess the term"**. Cut along the lines.

Loop cards (dominoes)

A closed loop of **16 cards**. Each card has an **ANSWER** (top) and a **CLUE** (bottom) pointing to the answer on another card. Deal the cards out; one student reads their clue, whoever holds the matching answer reads it and then their own clue, and so on until the loop returns to the start.

How to play 1. Cut out and shuffle all 16 cards; deal them to the class (some students may hold more than one). 2. The teacher (or the student holding Card 1) reads the **bottom clue** of Card 1 aloud. 3. Whoever holds the card whose **top ANSWER** matches reads "I have [answer]", then reads their own **bottom clue**. 4. Continue until the chain loops back to Card 1 — if it closes with no cards left over, the loop is verified correct.

Card	TOP — Answer (read when your card is called)	BOTTOM — Clue (read this to find the next card)
1	**Recursion**	…a subroutine that calls itself; what is the condition that **stops** it called?
2	**Base case**	…if that stopping condition is missing or never reached, what **error** results?
3	**Stack overflow**	…that error happens when too many frames are pushed onto which structure?
4	**Call stack**	…the call stack tracks where variables can be accessed; what is that "region of visibility" called?
5	**Scope**	…a variable visible **only inside the subroutine** that declares it is which kind?
6	**Local variable**	…and a variable visible **throughout the whole program** is which kind?
7	**Global variable**	…a subroutine that performs a task and **returns a single value** is called what?
8	**Function**	…a subroutine that performs a task but **returns no value** is called what?
9	**Procedure**	…passing a **copy** of an argument so the original is unchanged is which method?
10	**By value**	…passing the **memory address** so the original can be changed is which method?
11	**By reference**	…bundling data + methods and keeping attributes **private** (data hiding) is which OOP principle?
12	**Encapsulation**	…a subclass **acquiring** the attributes and methods of a superclass is which OOP principle?
13	**Inheritance**	…the same method call **behaving differently** for different object types is which OOP principle?
14	**Polymorphism**	…building a solution step by step and, on a **dead end**, returning to the last choice — which method?
15	**Backtracking**	…a **rule of thumb** giving a good-enough (not guaranteed optimal) answer quickly — which method?
16	**Heuristic**	…a subroutine that **calls itself**, needing a base case and a recursive case — what is it? *(loops to Card 1)*

Verification: 1→2→3→4→5→6→7→8→9→10→11→12→13→14→15→16→1 — a single closed loop with no breaks.

Taboo cards

8 cards. Get your team to say the **TERM** without using any of the 4 **BANNED** words (or obvious variants).

How to play 1. The describer takes a card and gives clues so their team guesses the **TERM**. 2. They may **not** say the term itself or any of the 4 banned words (or close variants/abbreviations). 3. 1 point per correct guess in 60 seconds; an opponent watches for banned words (-1 if used).

#	TERM	BANNED words
1	**Recursion**	itself · base case · calls · stack
2	**Encapsulation**	private · data · methods · hiding
3	**Inheritance**	subclass · parent · acquires · superclass
4	**By reference**	address · copy · original · memory
5	**Heuristic**	rule of thumb · optimal · good-enough · quick
6	**Local variable**	subroutine · scope · global · inside
7	**Polymorphism**	method · override · different · object
8	**Backtracking**	dead end · undo · maze · try

Bingo

How to play 1. Each student draws a 3×3 grid and writes any **9** of the 20 terms from the word bank. 2. The caller reads a clue from the numbered list (in random order), without saying the term. 3. Students cross off the matching term. First to a line (or full house) shouts "Bingo!" — verify their crossed terms against the called clues.

Word bank (choose 9):

Sequence · Selection · Count-controlled iteration · Condition-controlled iteration · Recursion · Base case · Call stack · Stack overflow · Scope · Local variable · Global variable · Function · Procedure · By value · By reference · Class · Object · Encapsulation · Inheritance · Heuristic

Caller's clue list:

1. Instructions executed one after another, in order. → *Sequence*
2. Choosing between paths based on a condition (if / switch). → *Selection*
3. A loop repeating a fixed, known number of times (for…next). → *Count-controlled iteration*
4. A loop repeating until a condition changes (while / do…until). → *Condition-controlled iteration*
5. A subroutine that calls itself, with a base case and a recursive case. → *Recursion*
6. The condition that stops recursion — no further recursive call. → *Base case*
7. The structure of frames holding parameters, locals and return addresses for active calls. → *Call stack*
8. The error from too many pushed frames, e.g. a missing base case. → *Stack overflow*
9. The region of a program where a variable can be accessed. → *Scope*
10. A variable accessible only within the subroutine that declares it. → *Local variable*
11. A variable accessible throughout the whole program. → *Global variable*
12. A subroutine that performs a task and returns a single value. → *Function*
13. A subroutine that performs a task but returns no value. → *Procedure*
14. Passing a copy of an argument, so the original is unaffected. → *By value*
15. Passing the memory address, so the original can be changed. → *By reference*
16. A template/blueprint defining the attributes and methods of its objects. → *Class*
17. A specific instance of a class, created at run time. → *Object*

18. Bundling data + methods and keeping attributes private (data hiding). → *Encapsulation*
19. A subclass acquiring the attributes and methods of a superclass. → *Inheritance*
20. A rule of thumb giving a good-enough (not guaranteed optimal) answer quickly. → *Heuristic*

Quick-fire "Guess the term"

12 rapid clue → answer pairs. Read the clue; first correct answer scores. (Answers in **bold** — fold or cover before playing.)

How to play 1. Split into two teams. The caller reads a clue. 2. First team to call the correct term wins the point; pass on a wrong answer. 3. Run all 12; highest score wins.

#	Clue	Answer
1	A `while` loop tests its condition here, so the body may run zero times.	**At the start (pre-conditioned)**
2	A `do…until` loop guarantees the body runs at least this many times.	**Once (post-conditioned)**
3	Each recursive call pushes one of these onto the call stack.	**A stack frame**
4	The variable in a subroutine *definition* that receives an argument.	**Parameter**
5	The actual value supplied at the point of the *call*.	**Argument**
6	A local of the same name does this to a global inside its subroutine.	**Shadows it**
7	OCR's keyword for the constructor method that runs on object creation.	**new**
8	A subclass replacing an inherited method with the same name and parameters.	**Overriding**
9	Same method name but different parameter lists.	**Overloading**
10	Searching large data sets for hidden patterns, e.g. basket analysis.	**Data mining**
11	Splitting a process into stages so different items are processed at once.	**Pipelining**
12	Repeatedly splitting a problem into smaller instances, solving and combining.	**Divide and conquer**